

Estela Pérez Luque, PhD

Data Scientist

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PROFILE

Data Scientist with a PhD in Informatics and 6+ years of experience building data pipelines and analyzing complex, real-world datasets in automotive engineering. Skilled at designing reproducible workflows that transform experimental and simulation data into actionable insights, automating analysis tasks, and communicating results clearly to non-technical stakeholders. Core work combines Python and MATLAB-based data processing, time-series analysis, and predictive modelling across large, messy datasets. Drawn to Einride specifically because the challenge of scaling autonomous freight through data-driven systems is exactly the kind of meaningful, fast-moving problem I want to work on.

TECHNICAL SKILLS

Programming: MATLAB, Python, SQL (basic), C++ (basic)

Data & Analytics: End-to-end data pipelines, data cleaning and structuring, reproducible workflows, time-series analysis, model benchmarking, insight generation, and reporting

Domain: Automotive engineering, human factors, experimental data systems

Languages: English (C1), Spanish (Native), Swedish (B1-improving)

PROFESSIONAL EXPERIENCE

Research Engineer (Simulation & Data Science) - University of Skövde, Sweden (Onsite) Sep 2020 – Oct 2025

- Built end-to-end Python and MATLAB pipelines to ingest, clean, and benchmark predictive models against motion capture and in-vehicle datasets spanning up to 199 participants across multiple tasks and conditions, generating traceable evidence for model validation and engineering decisions at scale.
- Standardized model benchmarking workflows across datasets and configurations, replacing time-consuming manual processes with reusable pipelines and freeing analytical capacity for higher-value problem-solving.
- Applied multi-objective optimisation (NSGA-II) across 1,000+ vehicle interior configurations and 14 digital human models, transforming a manual, trial-and-error ergonomics process into a fully automated, data-driven workflow that delivered consistent results at a previously unachievable scale.
- Developed a multi-metric framework combining RMSE and DTW across 10 participants and 4 reach task types, uncovering up to 15-20% variation in model performance by evaluation approach and sharpening the rigour of model selection.
- Designed scenario-based simulation studies comparing simulated and real-world behaviour across multiple system configurations and user groups, reducing reliance on subjective assessment and improving consistency in early development stages.
- Collaborated with Volvo Cars, Volvo Trucks, Scania, and the SAFER ecosystem to align validation strategies with real-world engineering constraints, translating findings into actionable recommendations through 20+ peer-reviewed publications and 15+ international presentations.

Visiting Researcher - University of Antwerp, Belgium (Onsite)

Sep 2024 – Mar 2025

- Designed and executed motion capture and 4D body scanning protocols for 30+ participants, ensuring consistent data quality and reproducibility across experimental conditions.
- Built structured, reproducible data pipelines to standardize multi-site data collection and processing, increasing data consistency and comparability across institutions and supporting more robust cross-study validation.

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Co-founder and R&D Lead - Swe-Ex AB, Sweden (Startup)

Sep 2021 – Mar 2023

- Co-founded an early-stage startup developing wearable safety support equipment for assembly-line workers in industrial environments, focused on ergonomic risk reduction and injury prevention.
- Translated user needs and safety requirements into measurable design criteria, driving three validated prototype iterations from CAD concept to physical testing using 3D-printed, textile, and steel components.
- Through two years in a structured incubator program, developed practical business acumen in value proposition development, stakeholder communication, and prioritization under uncertainty, learning to connect technical feasibility with product direction and commercial viability.
- Operated in a fast-moving, resource-constrained environment where adaptability and evidence-based decisions were essential, building direct experience in the kind of high-pace, high-impact work that defines companies transforming an industry.

Research Assistant - University of Skövde, Sweden (Onsite)

Jun 2019 – Sep 2020

- Analyzed simulation data and produced decision-ready reports and visual outputs across 6+ industry projects, enabling stakeholders to act on ergonomic insights during early-stage product development.

INDEPENDENT PROJECTS | Data Science Portfolio

Nov 2025 – Present

- Built a personal data science portfolio applying Python-based ML methods, data pipelines, and analytical workflows to real datasets across multiple project areas.
- Developed a portfolio web application using LLM-assisted development tools (Claude Code & Chat GPT), gaining hands-on experience in rapid prototyping and iterative delivery with React, TypeScript, and Vite.

EDUCATION

PhD in Informatics, University of Skövde (Sweden)

Sep 2020 – Oct 2025

[PhD Thesis](#): Human Posture and Motion Prediction for Automotive Ergonomics Design

Focus: Computational modelling, simulation, human motion prediction, experimental validation, time-series

Master of Science in Intelligent Automation, University of Skövde (Sweden)

Aug 2018 – Jun 2019

Bachelor of Science in Mechanical Engineering, University of Skövde (Sweden)

Aug 2017 – Jun 2018

Degree in Mechanical Engineering, University of Málaga (Spain)

Aug 2012 – Jun 2018

RESEARCH PUBLICATIONS

Pérez Luque, E., Lee, S., Högberg, D., Yang, J., & Lamb, M. (2026). Predicting Human Upper Extremity Reaching Motions: Comparison of Optimization-Based Method and Heuristic Method. International Journal of Human-Computer Interaction. DOI: 10.1080/10447318.2026.2632154

Full publication list available at: [Google Scholar – Estela Pérez Luque](#)